

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459938.7899 +/- 0.00084 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1333 +/- 0.0067
Transit depth (Rp/Rs)^2	1.78 +/- 0.18 [%]
Orbital Inclination (inc)	80.74 +/- 1.5
Airmass coefficient 1 (a1)	1.1513 +/- 0.0061
Airmass coefficient 2 (a2)	-0.13 +/- 0.012
Scatter in the residuals of the lightcurve fit is	0.53 %
Best Comparison Star	None
Optimal Aperture	4.6
Optimal Annulus	9.08
Transit Duration (day)	0.1205 +/- 0.0042

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1333 ± 0.0067 vs 0.1178 (deviation 2.31σ)
Duration fit vs published	0.1205 d vs 0.125 d (deviation 1.07σ)
Mid-time O–C	1573.4 min
Depth SNR	19.9
Residual scatter	0.53 %

Light curve

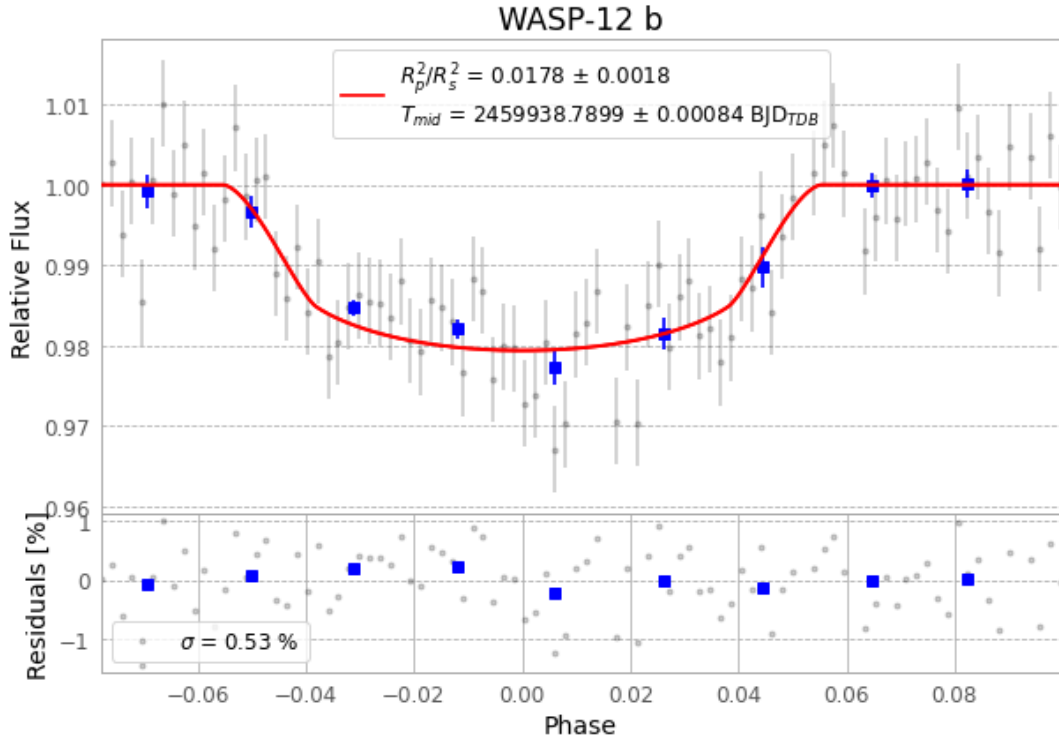


Figure 1 — FinalLightCurve_WASP-12 b_2022-12-24.png (SHA-256 bf3c2693f1b09c02...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [420, 110], 2 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 9a13f58331c18741...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

94 FITS frames (62 MB), WASP-12221225044515.FITS → WASP-12221225092415.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-221224.log (1eee39500fde...), FinalLightCurve_WASP-12 b_2022-12-24.png (bf3c2693f1b0...), FinalLightCurve_WASP-12 b_2022-12-24.csv (f77af6cdb5ab...)

Compute resources

Wall time 105.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 10:33Z.